

Dinul Iman

Full-stack Developer · Jakarta, Indonesia

Full-stack developer focused on the backend, with a love for working with data — from payroll systems to ETL pipelines.

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ABOUT

Hi! I'm Dinul Iman, an Information Technology graduate from Universitas Sumatera Utara focused on full-stack development, with my strongest experience in backend and databases. Day to day I work with PHP, TypeScript, Node.js and relational databases such as PostgreSQL and Oracle.

My most notable work is the Payroll Management System I worked on for PT PELNI as a programmer at PT Solusi Integrasi Utama, a PELNI vendor. It serves around 4,800–5,000 employees — from PPh 21 income tax, BPJS, pension and cooperative deductions to journal integration with Oracle. A favourite achievement: cutting the payroll run from more than 10 minutes to under 1 minute by optimizing stored procedures.

I enjoy work that sits close to data: ETL/ELT, data warehousing and automation. Lately I've also been actively exploring AI coding agents such as Claude Code, Codex and MCP. Outside work I build open portfolio projects: document extraction, Text-to-SQL and RAG with local LLMs, a Midtrans checkout that survives misbehaving webhooks, data pipelines with Airflow, Kafka, Spark and AWS/Terraform, and a forecast model that is monitored and retrained in production.

TARGET ROLES

Backend Developer / Engineer · **Full-stack Developer** · **Software Engineer** · **Data Engineer** · **HRIS / Payroll Technology** · **Freelance: backend, data & AI automation**

EXPERIENCE

Programmer — PT Solusi Integrasi Utama (PT PELNI vendor) Aug 2025 — Present

- PT PELNI Payroll Management System (±4,800–5,000 employees): development with PHP/CodeIgniter and PostgreSQL.
- Payroll calculation and processing: PPh 21 income tax, BPJS, pension, cooperative, receivables and salary proration.
- Worked on the data integration Oracle → PostgreSQL → payroll → Oracle journal.
- Optimized payroll stored procedures, cutting a run that took >10 minutes to <1 minute.
- Built payslips and payroll reports with JasperReports/JasperServer.
- PT PELNI ticketing application: backend development with NestJS and PostgreSQL, including the shipping insurance module.

- Used Docker / Rancher Desktop for application environments.

Data Structures & Algorithms Lab Assistant — Universitas Sumatera Utara

Jan 2021 — Jun 2021

- Teaching assistant for the Data Structures & Algorithms lab course.

Database Lab Assistant — Universitas Sumatera Utara Jun 2020 — Jan 2021

- Teaching assistant for the Database lab course.

EDUCATION

B.Sc. Information Technology — Universitas Sumatera Utara (USU)
GPA 3.50 · Lab assistant for Databases and for Data Structures & Algorithms.

SKILLS

PHP & CodeIgniter	★★★★★	Laravel	★★★★☆
Yii2	★★★☆☆	Node.js & NestJS	★★★★☆
TypeScript	★★★★☆	Python (FastAPI, Django)	★★★☆☆
Go & Gin	★★★☆☆	JavaScript	★★★★☆
React & Next.js	★★★☆☆	PostgreSQL	★★★★★
Oracle	★★★★☆	MySQL	★★★★☆
MongoDB	★★★☆☆	ETL & Data Integration	★★★★☆
JasperReports / JasperServer	★★★★☆	Tableau	★★★☆☆
BigQuery	★★★☆☆	Airflow & dbt	★★★☆☆
Spark & Kafka	★★★☆☆	Docker / Rancher	★★★☆☆
Payroll & HRIS	★★★★★	LLM & AI Integration	★★★☆☆
n8n & Workflow Automation	★★★☆☆	Web Scraping	★★★★☆
Payment Gateway	★★★☆☆		

PROJECTS

Payroll Management System for PT PELNI
Payroll system for ±4,800–5,000 employees: PPh 21 income tax, BPJS, pension, cooperative, receivables and proration; Oracle ↔ PostgreSQL integration; payslips & reports via JasperServer. Stored procedure optimization cut the run from >10 minutes to <1 minute. Built as a programmer at PT Solusi Integrasi Utama.

PHP · CodeIgniter · PostgreSQL · Oracle · JasperServer · Docker

Ticketing Application for PT PELNI
Backend for a ticketing application, including a shipping insurance module. Built at PT Solusi Integrasi Utama.

NestJS TypeScript PostgreSQL

Data Engineering Portfolio link

Three end-to-end data pipelines built like small production systems, with CI: batch ELT of weather data for 10 Indonesian cities (Airflow, MinIO, PostgreSQL, dbt, Metabase), real-time e-commerce order streaming (Kafka, Spark Structured Streaming with watermarks, dead-letter queue, Grafana), and an NYC Taxi bronze → silver → gold lakehouse (PySpark, Delta Lake, quality gates, time travel).

Python Airflow dbt Spark Kafka Delta Lake PostgreSQL Docker

Retail Analytics — 1 Juta Transaksi Nyata link

Analysis of the UCI Online Retail II dataset (1,067,371 lines): tested data cleaning (including an 80,995-unit order entered by mistake), SQL analysis (cohorts, RFM, market basket), a Metabase dashboard built from code, and a findings report: 25% of customers bring 70% of revenue.

Python Pandas PostgreSQL SQL Metabase

Otomasi Laporan Excel link

Sales spreadsheets from 6 branches, each in a different layout, are read automatically, cleaned and validated, loaded into PostgreSQL, then an Excel + PDF report is emailed to management and each branch receives the rows it needs to fix.

Python openpyxl PostgreSQL ReportLab RapidFuzz

Quake Warehouse — ELT Inkremental dengan dbt link

ELT from the live USGS earthquake API: landing zone, append-only raw that keeps every revision, incremental dbt models on DuckDB (BigQuery target ready), 23 data tests per build. Three bugs that only real data reveals were found and covered by tests.

Python dbt DuckDB SQL GitHub Actions

CDC Pipeline — Debezium + Kafka link

Change data capture from an operational PostgreSQL into an analytics database in near real time (p50 lag about 1 s). Idempotent by LSN, full change history, automatic schema evolution. Chaos-tested (consumer crash, Debezium restart, new column) and verified row by row in CI.

Debezium Kafka PostgreSQL Python Docker

Demand Forecasting — 12.196 Produk Walmart link

Daily sales forecasts 28 days ahead from the real M5 competition data. One LightGBM model for every product, leakage-tested features, a rolling backtest, then a single run on the competition's hidden period: 30% lower error than the baseline and calibrated 80% intervals (78.9% coverage). Streamlit dashboard included.

Python LightGBM Pandas Streamlit

Analisis A/B Test link

A real 90,000-player experiment (Cookie Cats) analysed properly: it finds a group-split imbalance that most analyses miss, adds sensitivity checks and a Bayesian view, plus

simulations showing why peeking at results is dangerous (5% → 27% false positives) and what CUPED saves.

Python SciPy Statistik Pandas

Event Analytics dengan ClickHouse link

Self-hosted product analytics in the style of Google Analytics: collector API → ClickHouse → Grafana. 20 million events, funnels, retention, A/B results, and detection of an app release that broke payments. Every number is checked against ground truth; load-tested at 14,000 events/s with nothing lost or duplicated.

ClickHouse FastAPI Grafana Python Docker

Data Quality Monitor link

Data quality monitoring for a warehouse of USGS earthquake data: freshness, volume, distribution drift (PSI), schema, reconciliation, Slack alerts on status changes, and an HTML report. Tested with 8 injected incidents and replayed over 76 days of real data without a single false failure.

Python DuckDB dbt SQL

Government Regulation Q&A (RAG) link

Q&A over Indonesian regulations PP 35/2021 and PP 36/2021 with their amendments (official PDFs); every answer cites its article. Hybrid search with pgvector and full-text, a local LLM via Ollama. Evaluated on 30 questions: 25/25 facts correct, 5/5 out-of-scope questions refused, zero invented citations.

Python PostgreSQL pgvector Ollama FastAPI

API → Data Warehouse Pipeline link

Incremental loads from a shop API (pagination, 429 rate limits, 500 errors, late-arriving changes) and real ECB exchange rates with dlt into DuckDB, modelled with dbt (28 tests). The result is checked against ground truth; three real bugs found and fixed.

Python dlt dbt DuckDB

Serverless AWS Pipeline with Terraform link

CSV files land in S3 and a Lambda validates every row: clean data to a date-partitioned lake, bad rows quarantined, daily counters in DynamoDB (no double counting on retries), failures to an SQS dead-letter queue, a daily report via EventBridge. All in Terraform with least-privilege IAM; 29 tests run on LocalStack and in CI.

AWS Terraform Python LocalStack

MLOps: Forecast Model in Production link

A LightGBM model for 3,049 Walmart products served through an API with an MLflow registry, monitored (data quality, drift, accuracy, bias) and retrained automatically behind a gate. A year of operations replayed: retrained models were 4–6% more accurate per product, and a prices-in-cents incident was caught the same day. Grafana + Prometheus dashboards.

Python LightGBM MLflow FastAPI Grafana

Naver SmartStore Scraper API link

REST API that extracts product data from Naver SmartStore pages using a stealth headless browser, rotating proxies & fingerprints, throttling and caching. The scraping backend can be swapped via an env var with no code changes.

TypeScript Node.js Playwright

AI Document Extractor link

Turns invoice PDFs into structured data in PostgreSQL with a local LLM (Ollama), so documents never leave the server. The model only reads; dates and currency are normalised in code, then the numbers are checked (qty × price, subtotal, tax, total). Anything that doesn't add up goes to a review queue. 100% accuracy on the test set (60/60 fields, 20/20 line items).

Python FastAPI Ollama PostgreSQL Pydantic

Text-to-SQL Agent link

Ask a PostgreSQL database in plain language (English/Indonesian): a local LLM writes the SQL, then a sqlglot guard and a read-only database role make sure it can only read. Results come back as a table and a chart. 20/24 correct on the test set, and 6/6 hostile requests refused without touching the data.

Python FastAPI Ollama sqlglot PostgreSQL

AI Ticket Triage with n8n link

n8n workflow that reads customer messages (Indonesian/English), classifies category & priority with a local LLM, drafts a reply, stores the ticket in PostgreSQL and alerts Slack/Discord when it's urgent. Plus a daily digest.

n8n Ollama PostgreSQL JavaScript Docker

Price Tracker link

Async scraper that collects 1,000 products in about 60 seconds with retries and rate limiting, keeps price history in PostgreSQL, detects price/stock changes, sends alerts and shows a Streamlit dashboard.

Python httpx PostgreSQL Streamlit pytest

PostgreSQL Payroll Optimization Case Study link

A real payroll problem rebuilt with synthetic data for 5,000 employees: from a 28-second row-by-row loop to 0.2 seconds (140x) using indexes, caching outside the loop and a set-based version, with results verified identical row by row.

PostgreSQL PL/pgSQL Docker

Leave Management API link

Employee leave REST API in Laravel: working-day calculation (weekends & holidays excluded), per-type balances, pending requests that reserve balance, manager approval, and row locking so parallel requests stay safe. 20 tests run on SQLite and PostgreSQL in CI.

PHP Laravel Sanctum PostgreSQL PHPUnit GitHub Actions

GoBack — E-Commerce Platform link

Full-stack e-commerce app: Go backend with JWT auth, product & stock management, addresses, Midtrans payments and an admin dashboard; React frontend.

Go · Gin · GORM · PostgreSQL · React · JWT

Payment Webhooks (Midtrans) in Go link

Midtrans checkout that survives forged, duplicate, out-of-order and lost webhooks: signature and amount checks, database-level idempotency, a status machine and reconciliation via the status API. Ships with a local Midtrans simulator; every scenario runs in CI.

Go · Gin · PostgreSQL · Midtrans · Docker

Job Market Analytics

Job market analysis platform: job-posting scraping, automatic skill extraction, in-demand skills, salary statistics by location & level, and career path recommendations.

Python · Flask · Vue.js

Botto — RAG Chatbot link

Chatbot for asking questions about your own documents (PDF, Word, TXT) using Retrieval-Augmented Generation with streaming responses. The whole stack is free.

Next.js · TypeScript · LangChain.js · ChromaDB · Groq (Llama 3.1)

Toko Syuhada — Bicycle Shop Website link

Bicycle catalog & sales website with category filters, product search, detailed specs and articles.

Next.js · TypeScript · Prisma · PostgreSQL · Tailwind CSS

Machine Learning Experiments link

A collection of notebooks: an LSTM crypto Telegram chatbot (thesis), crypto price forecasting with LSTM, sentiment analysis, a seq2seq chatbot, and data analysis with Pandas and PySpark (car prices, ad campaigns, diabetes).

Python · TensorFlow/Keras · Pandas · PySpark

Apple or Tomato — Real-time Image Classifier link

Tells apples from tomatoes straight from a webcam: MobileNetV2 transfer learning with data augmentation, and an OpenCV app that shows the prediction and its confidence in real time.

Python · TensorFlow/Keras · MobileNetV2 · OpenCV

CV Quest link

An interactive CV as a browser RPG — yes, the game you're playing right now!

JavaScript · Canvas · HTML/CSS

INTERESTS

AI & Data · ETL/ELT & Data Warehousing · TypeScript · AI Coding Agents · Claude Code · Codex · MCP · Automation